Natural Language Processing - Lecture 5: Machine Translation (22 Oct)

Slide 1: Machine Translation

This title page identifies the lecture as Natural Language Processing: Machine Translation, taught by Miryam de Lhoneux at KU Leuven.

Slide 2: Lecture Outline

The lecture moves from the task definition and supervised training to decoding, evaluation, current performance, low-resource languages, gender bias, and open challenges.

Slide 3: Machine Translation Definition

Machine translation maps text from a source language to a target language. The English-to-Dutch example shows that the goal is a translated sentence, not a word-for-word gloss.

Slide 4: Information Access

Machine translation improves access to material written in languages a reader may not know. The slide frames information access as one practical reason to build MT systems.

Slide 5: Computer-Assisted Translation

Computer-assisted translation uses MT to support a human translator. A draft can speed up work, while the person checks meaning, terminology, and style.

Slide 6: Speech-to-Speech Translation

Speech-to-speech translation links speech recognition, translation, and speech output so speakers of different languages can interact more directly.

Slide 7: Image Translation

Image translation can recognize text in a scene and render it in another language. This is useful in travel, but accuracy depends on recognizing the text and preserving its context.

Slide 8: Word Order Differences

Languages differ in word order: English commonly uses subject-verb-object, Japanese often places the object before the verb, and Arabic commonly places the verb first. A translator must reorder constituents while preserving who did what to whom.

Slide 9: Lexical Ambiguity: Bank

The English word bank can map to different Dutch words depending on meaning, including a financial institution, a river edge, or a verb. Translation therefore requires word-sense selection from context.

Slide 10: Morphological Differences

The dog example highlights that languages divide lexical and grammatical categories differently. French forms can mark grammatical gender where English uses one noun; translation must preserve the intended referent and follow target-language morphology.

Slide 11: Lexical Gaps

A lexical gap occurs when a language lacks a single word with the same meaning and scope as a source word. The German examples require a phrase or explanation in English, so a translator must preserve the concept rather than force a one-word match.

Slide 12: Pro-Drop Languages

Spanish can omit a subject pronoun when verb morphology and context make the subject recoverable. English normally requires an explicit subject, so MT must infer and express information that is absent from the source string.

Slide 13: Supervised MT

This outline marks the transition to supervised machine translation, where the system learns from parallel examples.

Slide 14: Parallel Training Corpus

A parallel corpus pairs source and target sentences. The Santa-and-elves example shows how aligned sentence pairs provide supervised examples for training an MT model.

Slide 15: Sentence Alignment

Sentence alignment links corresponding sentences in two versions of a text. It is a preparatory step because training examples need source and target units that express the same content.

Slide 16: Non-One-to-One Alignment

The Little Prince example shows that translations do not always align sentence by sentence. A sentence may split, merge, or correspond to a span of different length in the other language.

Slide 17: Alignment Methods

Alignment can score candidate spans using length or multilingual embedding similarity, then use dynamic programming to select a consistent sequence of span pairs. The slide’s method allows groups of up to two sentences.

Slide 18: Conditional Sequence Modeling

A supervised neural MT model receives tokenized source and target sequences and learns P(target sequence | source sequence). Training raises the probability of the reference target given its source.

Slide 19: Encoder-Decoder Transformer

The encoder-decoder Transformer represents the source with an encoder and generates the target with a decoder. The decoder conditions on the encoded source while producing target tokens.

Slide 20: Self-Attention Review

Self-attention lets each token representation use information from other positions in the same sequence. MT uses this within the encoder and decoder to build contextual representations.

Slide 21: Cross-Attention

Cross-attention lets decoder positions consult the encoded source. In the diagram, the keys and values come from encoder outputs, allowing each generated target token to use relevant source information.

Slide 22: Encoder-Decoder Transformer Blocks

The encoder and decoder are stacks of Transformer blocks. Encoder self-attention processes the source; decoder self-attention uses prior target tokens, and cross-attention links the decoder to the source representation.

Slide 23: Attention Weights in Translation

The attention visualization links positions in a French source sentence to positions used while translating into English. Attention weights show internal information flow; they are not guaranteed to be human-readable word alignments.

Slide 24: Training Objective

Training predicts the next target token and uses cross-entropy loss. Teacher forcing supplies the correct earlier target tokens during training, while inference must condition on the model's own generated history.

Slide 25: Beam Search

This outline marks the decoding section. The trained model assigns probabilities to continuations; a decoding algorithm selects one output sequence.

Slide 26: Greedy Decoding

Greedy decoding selects the highest-probability next token at each step. A locally best token can lead to a sequence with lower total probability than another path, motivating beam search.

Slide 27: Search Tree for Language Modeling

The search tree represents alternative token sequences. Each branch adds a token, and a path corresponds to one partial translation.

Slide 28: Beam Width

Beam search keeps the best k partial sequences at each step instead of exploring every sequence. The beam width k controls the memory and search effort; pruning can discard a path that would later become best.

Slide 29: Beam Search with k = 2

With k = 2, the decoder carries two candidate sequences forward at each step. The illustration shows a bounded search, not an exhaustive guarantee of the globally best sequence.

Slide 30: Evaluation

This outline marks the evaluation section, which compares machine-generated translations with human judgments or reference translations.

Slide 31: Adequacy and Fluency

Adequacy asks whether the translation preserves the source meaning. Fluency asks whether it reads naturally and grammatically in the target language; these qualities can diverge.

Slide 32: Human Evaluation

Human evaluation can rate fluency and adequacy, rank candidate translations, or post-edit a draft and measure the effort. It assesses quality directly but takes time and trained reviewers.

Slide 33: Automatic Evaluation

Automatic metrics compare a candidate with one or more human reference translations. The lecture groups them into surface-overlap measures such as chrF and BLEU, and embedding-based measures.

Slide 34: chrF Metric

chrF compares character n-gram overlap between a hypothesis and reference. It averages precision and recall over character n-gram lengths up to k, which can give partial credit for morphological similarity.

Slide 35: chrF Formula

chrF combines character n-gram precision P and recall R with a beta-weighted F-score: $\text{chrF}\beta = (1 + \beta^2)PR / (\beta^2P + R)$. β controls the relative weight of recall; β = 1 gives equal weight, while larger β favors recall.

Slide 36: chrF Example: Precision

In the example, k = 2 and β = 2. The hypothesis has 17 unigrams and 16 bigrams; the reference has 18 unigrams and 17 bigrams. There are 17 matching unigrams and 13 matching bigrams, giving character precision (1 + 13/16) / 2 = 0.906. The comma counts as a character.

Slide 37: chrF Example: Recall

The recall calculation uses the reference as denominator: unigram recall is 17/18 = 0.944 and bigram recall is 13/17 = 0.765. Their average is 0.855.

Slide 38: chrF Example: F-score

Combining the example's P = 0.906 and R = 0.855 with β = 2 gives $\text{chrF}\beta \approx 0.86$. This is an overlap score for this reference and candidate, not a direct measure of meaning.

Slide 39: BLEU Metric

BLEU measures modified word n-gram precision for n-grams of lengths one through four and applies a brevity penalty when a candidate is too short. Corpus aggregation makes it useful for comparisons, but the score depends on reference wording.

Slide 40: BLEU N-gram Overlap

The illustration aligns n-grams between a reference and a candidate with different phrasing. Matching spans contribute overlap; paraphrased spans can fail to match even when they convey similar meaning.

Slide 41: Limits of chrF and BLEU

chrF and BLEU do not directly capture synonymy, meaning, word order beyond their n-gram patterns, or discourse across sentences. A fluent and adequate paraphrase can score poorly against one reference.

Slide 42: Embedding-Based Evaluation

Embedding-based metrics represent the candidate and reference in contextual vector spaces. A learned evaluator can predict human ratings or compare semantic representations, aiming to recognize similarity beyond exact word overlap.

Slide 43: BERTScore

BERTScore matches each reference token to its most similar candidate token and each candidate token to its most similar reference token. Recall averages the best similarity per reference token; precision averages the best similarity per candidate token.

Slide 44: State of the Art and Open Challenges

This outline introduces the final part on current MT quality and open challenges.

Slide 45: Translation Quality Tiers

The slide uses human-targeted edit rate (HTER) to describe how much post-editing a translation needs. The listed tiers move from publishable with little editing to triagable output useful only for locating relevant material; they are illustrative bands, not universal quality thresholds.

Slide 46: Quality by Language Pair and Domain

The examples show that MT quality varies by language pair, domain, and text type. The table reports assessments for the listed examples; it is not a permanent ranking of those languages or a current benchmark.

Slide 47: Low-Resource Machine Translation

Neural MT depends on large parallel corpora, which exist for only a fraction of language pairs. This data imbalance creates a low-resource problem even when methods work well for high-resource languages.

Slide 48: Backtranslation

Backtranslation augments training data by translating monolingual text to create synthetic source-target pairs. The synthetic pairs increase training material but can also introduce errors from the model that generated them.

Slide 49: Multilingual Models

A multilingual model handles multiple translation directions in one system. It shares parameters across language pairs and can reuse learned representations.

Slide 50: Cross-Lingual Transfer

Training one model on hundreds of language pairs can enable transfer between related languages, such as Spanish and Catalan. The slide reports competitive results for some pairs, not a guarantee for every language.

Slide 51: Low-Resource Challenges

Low-resource translation remains difficult because models need data and language properties such as rich morphology or different word order increase the challenge. The slide's list names obstacles rather than offering a single solution.

Slide 52: AmericasNLP Shared Task

The AmericasNLP shared task studies polysynthetic languages. The reported results show that baselines can be hard to beat and that LLM help depends on having sufficient target-side resources; retain this as a result from that task, not a universal rule.

Slide 53: Grammatical Gender Across Languages

Some languages mark grammatical gender on nouns or other words, while others such as English, Turkish, Basque, and Finnish do not use the same grammatical-gender system. Translation may need to add distinctions not marked in the source.

Slide 54: Gender Ambiguity in Translation

The English sentence "The doctor asked for help" does not specify the doctor's gender. Spanish requires a gendered noun form in the examples, so both la doctora and el doctor can be grammatical translations without more context.

Slide 55: Gendered Translation Choices

The example shows a translation system choosing one gendered form where the English source leaves gender unspecified. A single output can hide ambiguity; systems may default to a common or stereotyped association.

Slide 56: Gender Bias Example

This screenshot documents an earlier MT output that selected a gendered translation. It is evidence for the lecture's bias discussion at that time, not a claim about every current system.

Slide 57: Gender in Context

Context can determine which person a gendered expression refers to, and the target language may require an explicit gender choice. The example illustrates how an MT system can still make a stereotyped or unsupported choice.

Slide 58: Gender Bias Example

This screenshot continues the gender-bias examples. Evaluate the translated roles against the source context; the example shows why a plausible sentence can still encode the wrong gender assignment.

Slide 59: Gender Bias Evaluation

A gender-bias evaluation can create matched stereotypical and non-stereotypical examples, test the system on both, and compare gender translation accuracy. Balanced examples help separate model behavior from the wording of one sentence.

Slide 60: Non-Stereotypical Example

This non-stereotypical example uses a female doctor. It tests whether the system follows the source evidence instead of defaulting to a gender stereotype.

Slide 61: Stereotypical Example

This stereotypical example uses a male doctor. Comparing it with the matched non-stereotypical case helps reveal whether translation accuracy depends on the gender-role association.

Slide 62: Gender Translation Accuracy Results

The chart compares stereotypical and non-stereotypical translation accuracy for Google Translate across eight languages and displays their gap. The visible differences show that performance varies by language and example type; avoid treating this single evaluation as a general ranking of systems.

Slide 63: Gender Bias Mitigation

Gender bias remains an open MT challenge. The slide mentions controlled text generation as one mitigation approach and points to the Gender Inclusive Translation Technologies workshop; neither removes the need to measure performance on varied examples.

Slide 64: Open Challenges

The open-challenge list includes low-resource MT, gender bias, document-level context, decoding efficiency, rare words, and domain shift. These challenges concern coverage, context, search cost, and generalization.

Slide 65: Summary

This outline marks the lecture’s closing summary.

Slide 66: Lecture Summary

The summary recaps encoder-decoder neural MT, beam search, automatic evaluation and its limits, variation by language and domain, low-resource strategies, gender bias, and open challenges.

Slide 67: Course Reminder

Course reminder: the next meeting is Friday at 10:30. This is logistical information, not a machine-translation concept.

Slide 68: Acknowledgments

The slide acknowledges sources for the quality-tier and gender-bias sections. It credits the materials that informed those examples and directs students to Philipp Koehn’s MT course for further study.

Slide 69: References I

Reference list, part one. These citations document sources for multilingual MT, evaluation, data, and applications; retain them when following up on the examples.

Slide 70: References II

Reference list, part two. These citations continue the source bibliography for the lecture’s methods and open-challenge material.

Key points to remember

Supervised neural MT learns a conditional target-sequence distribution from parallel text and decodes a translation with methods such as beam search. chrF and BLEU compare surface overlap, while embedding-based metrics capture some semantic similarity; none replaces human judgments of adequacy and fluency. Quality varies by language pair and domain, with low-resource data and gender bias among the lecture’s open challenges.